

How to Run the Project

Campus Services Portal | Local development guide and user journey

Group A2 G1

Prerequisites

Install the .NET 8 SDK and a local Microsoft SQL Server instance. SQL Server LocalDB or SQL Server Express is suitable for development. An internet connection is required the first time NuGet packages are restored. A SendGrid account is needed only to test email verification, resend verification, or password reset.

Required Local Configuration

The repository does not include secrets. From the project folder, initialize user secrets if needed and set the following values. The connection string below is an example for LocalDB; change it if you use another SQL Server instance.

```
dotnet user-secrets init
dotnet user-secrets set "ConnectionStrings:DefaultConnection"
"Server=(localdb)\MSSQLLocalDB;Database=CampusServicePortalDb;Trusted_Connection=True;Trust
ServerCertificate=True"
dotnet user-secrets set "Jwt:Key" "replace-with-a-long-random-development-key"
dotnet user-secrets set "Jwt:Issuer" "CampusServicePortal"
dotnet user-secrets set "Jwt:Audience" "CampusServicePortalClient"
dotnet user-secrets set "Jwt:DurationInMinutes" "60"
```

For email functionality, additionally set SendGrid:ApiKey, SendGrid:FromEmail, SendGrid:FromName, and AppSettings:FrontendBaseUrl. Do not commit these secrets.

Start the Application

1. Open PowerShell or a terminal in the project root containing CampusServicePortal.csproj.
2. Restore packages: dotnet restore
3. Build the project: dotnet build
4. Run in Development mode: dotnet run --launch-profile http
5. On first launch, Entity Framework Core applies the included migrations and the application seeds the demo users and sample data automatically.

6. Open <http://localhost:5134> in a browser. Swagger API documentation is available at <http://localhost:5134/swagger> while running in Development.

Test Login Credentials

Role	Login identifier	Password
Administrator	Username: admin	Admin@123
Student	Email: student1@university.edu	Student@123
Student	Email: bthilaharaj@gmail.com	Jochim#2001

At the login screen, choose the correct role. The Administrator account uses its username, while the Student account uses its email address. These are seeded development accounts, not production credentials.

Frontend URL : <https://localhost:7133>

Troubleshooting

If startup reports a database connection error, check that SQL Server or LocalDB is installed and that ConnectionStrings:DefaultConnection points to it. If registration or reset email fails, confirm the SendGrid settings and verified sender. If port 5134 is unavailable, use the URL printed by dotnet run.

User Journey

Student Journey

7. Open the portal and choose Register or Login.
8. For registration, enter a valid university index number. The system checks the student master list and pre-fills the available student details.
9. Enter contact details, email address, and password, then verify the email address using the verification link. (After verify your mail the verification link send in your spam mail because of domain(gmail.com)).
10. Log in as a Student and open the student dashboard.
11. Use campus services: maintain the profile, apply for hostel accommodation, reserve a lab seat, submit complaints, view fees, register for events, request certificates, and view notifications.
12. Track request status and log out when finished.

Administrator Journey

13. Open the portal, select the Admin role, and log in with the administrator username and password.
14. Open the administration dashboard to review system summaries and pending work.
15. Manage student records and master data, including hostels and rooms, labs, complaint categories, fee types, venues, events, and certificate types.
16. Review requests and applications, update their status, and create notifications for students where needed.
17. Log out securely after administration is complete.